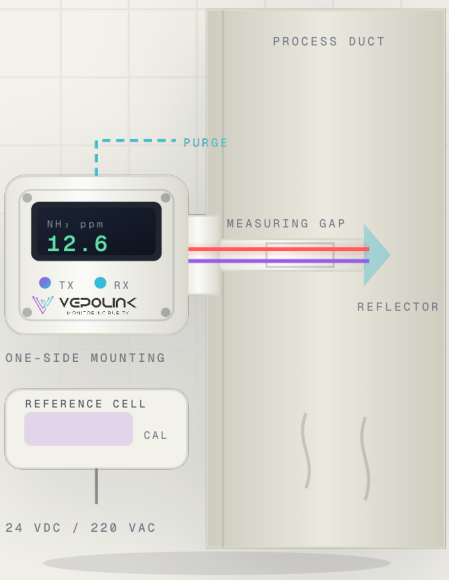
VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × GAS DETECTION

IN-SITU TDLAS · ONE-SIDE MOUNTING

In-Situ TDLAS Gas Analyzer *One flange, one second*

A high-precision tunable laser analyzer that scans the absorption line of the target gas and reads concentration from the second harmonic signal — mounted on a single flange, calibrated online, with nothing extracted from the process.



<1 s
RESPONSE TIME

1 %
REPEATABILITY

1 side
MOUNTING, NO PATH ALIGNMENT

TDLAS SECOND HARMONIC | **ONLINE** REFERENCE CELL | **24 VDC** / 220 VAC

<1s	1%	1% F.S.	0
RESPONSE TIME	REPEATABILITY	LINEARITY	SAMPLE CONDITIONING

01 OVERVIEW

In-situ precision, *installed on one side.*

A highly precise laser gas analyzer based on Tunable Diode Laser Absorption Spectroscopy. A tunable semiconductor laser scans the absorption line of the measured gas, and concentration is read from the second harmonic signal.

One-side mounting simplifies installation compared with cross-stack systems and removes the need for optical path adjustment: a single flange carries the probe, and the beam returns from a reflector across the measuring gap. An integral reference gas cell allows online calibration without disassembly.

AT A GLANCE

- TDLAS
- SECOND HARMONIC
- ONE-SIDE MOUNTING
- NON-CONTACT OPTICAL
- REFERENCE GAS CELL
- ONLINE CALIBRATION
- NO CONDITIONING
- 24 VDC / 220 VAC

· PRODUCT FEATURES

Second-Harmonic Detection

Wavelength modulation and 2f demodulation pull a small absorption out of a noisy background, giving high precision at low concentration with stable zero.

One-Side Installation

A single flange carries the whole instrument. There is no opposed mounting to align and no complex optical path adjustment at commissioning or after thermal movement.

Reference Cell Calibration

An internal reference gas cell allows online calibration without disassembly — the instrument checks itself against a known line while the process keeps running.

Non-Contact, Low Maintenance

Nothing touches the sample and nothing is extracted: no conditioning, no adsorption, no blockage. Low drift and long service life keep the cost of ownership down.

02 MEASUREMENT PRINCIPLE

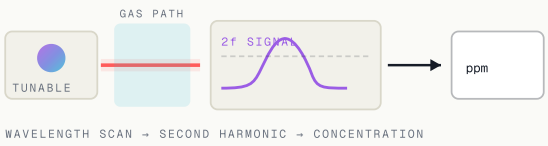
Scan the line, *read the harmonic.*

The tunable diode laser emits specific wavelengths through the gas path. The receiver processes the resulting signal and determines gas concentration from the second harmonic signal and the absorption characteristics of the target gas.

TDLAS

Line Scan & 2f Signal

TUNABLE SEMICONDUCTOR LASER · 2F DEMODULATION



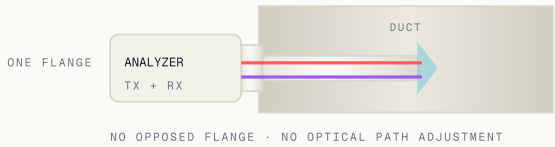
The laser wavelength is swept across a single absorption line while being modulated. Demodulating at twice the modulation frequency isolates the absorption from source and window losses.

`Concentration from the 2f amplitude`

INSTALLATION

One Flange, Folded Path

PROBE · IN-DUCT REFLECTOR · MEASURING GAP



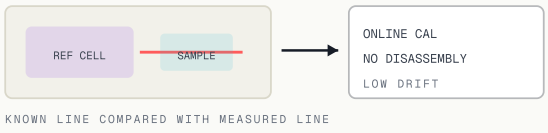
Transmitter and receiver sit in one head on one side of the duct; the beam crosses the measuring gap, returns from a reflector and is detected. No opposed flange, no alignment across the duct.

`Simpler than cross-stack`

SELECTIVITY

Single-Line Spectrum

REFERENCE GAS CELL · ONLINE CALIBRATION



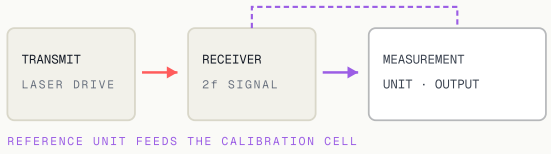
Working on one isolated spectral line reduces interference from background gases, and the internal reference cell provides a known line to calibrate against — online, without disassembly.

`Low drift, verified in place`

SYSTEM

Four Functional Units

TRANSMITTER · RECEIVER · REFERENCE · MEASUREMENT



The transmitter drives the laser, the receiver recovers the second harmonic, the reference unit supplies the calibration cell, and the measurement unit computes and outputs concentration.

`One integrated instrument`

03 SYSTEM & CONFIGURATION

Four units, *one measurement.*

The analyzer is configured to the gas and the duct. Measuring range is set to customer requirement — tell us the component, the expected concentration, the gap and the process conditions, and the configuration is confirmed at order.

UNIT	FUNCTION
Transmitter	Drives the tunable diode laser to emit light at specific wavelengths
Receiver	Receives and processes the optical signal to obtain the second harmonic signal
Reference Unit	Provides reference gas cell functionality for online calibration
Measurement Unit	Processes the measured signal and calculates the gas concentration

PROPERTY	CONFIGURATION
Installation	In-situ, one-side mounting on a single flange
Optical Path	Folded measuring gap; no path adjustment required
Sample Handling	None — no conditioning, no extraction
Calibration	Online, via reference gas cell, no disassembly

STEP 01

Flange

One flange on the duct wall carries the probe and the analyzer head; the reflector sits at the end of the measuring gap.

STEP 02

Power

24 VDC or 220 VAC to the head. No opposed unit to power, wire or align across the duct.

STEP 03

Purge

Industrial nitrogen or purified instrument air at 0.3–0.8 MPa keeps the optical surfaces clean in dusty service.

STEP 04

Verify

The reference gas cell provides an online calibration check against a known line — no disassembly, no downtime.

Because measurement is non-contact and in-situ, the failure modes of extractive systems simply do not apply: nothing adsorbs the sample on the way to the cell, nothing blocks, and nothing is damaged by condensate. Response is under a second, so the reading is usable for control and not only for record.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

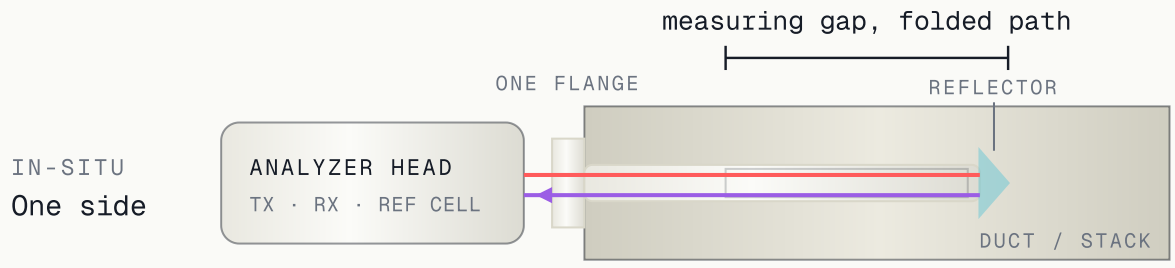
Sensor Technology	TDLAS tunable diode laser
Measurement Method	Non-contact optical
Measurement Type	In-situ
Signal Processing	Second harmonic (2f) of the scanned line
Measuring Range	As per customer requirement
Repeatability	1 %
Linearity	±1 % F.S.
Response Time	< 1 s
Selectivity	Single-line spectrum, low background interference

INSTALLATION · UTILITIES

Installation Method	In-situ, one-side mounting
Optical Path Adjustment	Not required
Sample Conditioning	Not required
Calibration	Online, reference gas cell, no disassembly
System Composition	Transmitter · receiver · reference unit · measurement unit
Power Supply	24 VDC or 220 VAC
Purge Gas	0.3-0.8 MPa industrial nitrogen or purified instrument air
Ambient Temperature	-20 °C to +60 °C
Service Life	Long, with low drift

INSTALLATION GEOMETRY

SINGLE FLANGE · PROBE AND REFLECTOR · FOLDED OPTICAL PATH



05 INSTALLATION & UPKEEP

One flange in, *years of service.*

> The simplification is real: **one flange instead of two opposed ones**, **no optical path adjustment** to make or to remake after the duct moves, and **no sampling system at all**. What remains is choosing a representative section, keeping the windows purged, and letting the reference cell prove the calibration online.

01 · LOCATION

Representative section

A straight run with well-mixed flow and stable conditions, clear of bends, dampers and air ingress.

02 · FLANGE

One side only

A single mounting flange carries the probe and head. No opposed flange to fabricate, wire or align.

03 · PROBE

Measuring gap

The probe defines the gap; the reflector returns the beam. Path length is set for the gas and the expected concentration.

04 · PURGE

N_2 or instrument air

0.3–0.8 MPa keeps the optical surfaces clean in dust-laden gas and protects them from condensate.

05 · CALIBRATION

Online reference cell

The internal cell provides a known absorption line for verification and adjustment without disassembly or process interruption.

06 · SERVICE

Clean and check

Periodic window cleaning and a reference check. There is no probe filter, heated line, pump or cooler to maintain.

ONE-SIDE VERSUS CROSS-DUCT

Both are TDLAS and both measure in place. A cross duct analyzer puts transmitter and receiver on opposite walls and suits very large ducts where a long optical channel is wanted. This one-side instrument folds the path onto a reflector, so a single flange, a single cable run and no cross-duct alignment are needed — usually the faster and cheaper installation where the gap is moderate.

COST OF OWNERSHIP

Low drift, a long service life, no consumable sampling components and online calibration together keep maintenance cost low. The instrument that needs no sample handling is the one that is still reading correctly a year after commissioning.

06 APPLICATIONS

Where it *works.*

Real-time concentration where the process is, for both the emission record and the control loop.

Continuous Emissions Monitoring	Industrial Process Gas Analysis	Environmental Compliance
Power Plants	Petrochemical Plants	Manufacturing Plants
Boiler & Furnace Ducts	Cement, Steel & Kilns	Waste Incineration & SCR

FAST ENOUGH TO CONTROL

A sub-second, in-situ reading closes loops that a minute-old extractive value cannot: excess-air trim on a burner, ammonia dosing on an SCR, safety interlocks on a process line. The measurement reflects the duct as it is, not as it was by the time the sample reached the cell.

SPECIFYING A MEASUREMENT POINT

We need the target gas and expected range, the duct dimension and available flange, the process temperature, pressure and dust loading, and the area classification. From that we confirm the measuring gap, the purge requirement, the mounting and the output configuration as one package.

07 FAQ

Frequently asked *questions.*

Q What is the In-Situ TDLAS Gas Analyzer?

A high-precision industrial gas analyzer using Tunable Diode Laser Absorption Spectroscopy for direct, in-situ gas concentration measurement.

Q How does it measure gases?

A tunable laser scans specific absorption lines and the analyzer calculates concentration from the second harmonic signal, without extracting a sample.

Q What are the key benefits?

Non-contact optical measurement, low drift, fast response of less than one second, and low maintenance requirements.

Q Is installation complicated?

No. The in-situ, one-side installation does not require complex optical path adjustment and is designed for reliable operation in industrial environments.

Q Where is it used?

Continuous emissions monitoring, industrial process gas analysis and environmental compliance in power, petrochemical and manufacturing plants.

Q How is it calibrated?

Online, using the internal reference gas cell. No disassembly is required and the process does not have to be interrupted.

Q What is the measuring range?

Configured as per customer requirement — set to the target gas, the measuring gap and the concentrations expected at the point.

Q Do background gases interfere?

Interference is minimised by single-line spectrum technology: the analyzer works on one isolated absorption line of the target gas.

Q What utilities does it need?

24 VDC or 220 VAC power, and 0.3–0.8 MPa industrial nitrogen or purified instrument air for purging. Ambient −20 °C to +60 °C.

Q How does it compare with a cross duct analyzer?

Both measure in situ by TDLAS. Cross duct suits very large ducts with a long optical channel; one-side mounting suits moderate gaps and is the simpler installation.

This brochure is general product information. Measuring range, measuring gap, mounting arrangement and performance figures are indicative and confirmed at order against the process conditions of the measurement point. Follow the user manual, site procedures and applicable standards for installation, purging, calibration and use. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

IN-SITU, IN UNDER A SECOND

One flange is all *it takes.*

Send us the gas, the duct and the range you need, and we'll confirm the measuring gap, purge, mounting and outputs as one package. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that specs, installs and commissions laser gas analyzers. No call centre.
+91 99538 86178 →

02 **Request a configuration** RECOMMENDED
Share gas, expected range, duct dimension and process conditions — we'll return gap, purge, mounting and output configuration.
sales@vepolink.com →

03 **Book a demonstration**
See the live 2f signal, an online reference-cell calibration and the output on a working analyzer.
Request a demo →